Exploring Electrical Energy and Power

The goal of this investigation is for you to develop some understanding of how electricity works and how energy transfers through electric circuits. The instructions are meant to guide your exploration so that you discover some key concepts. The goal here is learning, not completion, so work at a pace that helps you build understanding and feel free to work with your peers.

The first part of the assignment uses an online circuit simulator. Feel free to explore beyond the questions in the assignment and as always don't forget to ***record your questions***.

Setup the Simulator

- Open the simulator at https://bit.ly/dc-circuit-sim
- Click on the Intro option

Circuit 1

- Create a **1.5 Volt battery** in the simulator by dragging a battery onto the table. Clicking on the battery to bring up its options, and change the voltage to 1.5 V.
- Drag a light bulb onto the table.
- Use wires to connect the battery and bulb so that the light turns on.

Keep Circuit 1 on your virtual table for comparison and build Circuit 2.

Circuit 2

- Create **two 1.5 Volt batteries** as in Circuit 1.
- Connect the positive (+) end of one battery to the negative (–) end of the second battery.
- Connect the circuit so that the light bulb lights up.

Keep Circuit 1 and 2 on your virtual table for comparison and build Circuit 3.

Circuit 3

- Create a 9 Volt battery.
- Connect the circuit so that the light bulb lights up.

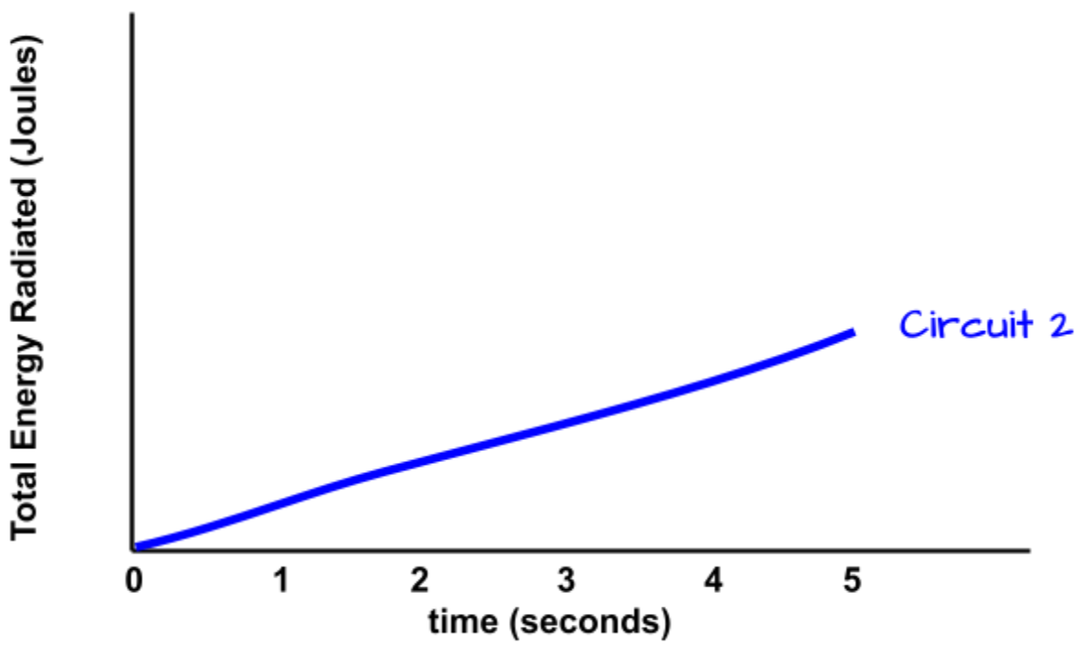
Don't close this browser window, you will come back to your circuits to take some measurements in a little while.

1. Take a screenshot of your circuits to add to your notes.
    a. Label the battery **voltage** for each circuit.
    b. Use the **Ammeter** tool to measure the **current** in **Amps (A)** at different points in each circuit and label the current on your screenshot.
    c. What do you notice about the relationship between the **voltage** and the **current**?
    d. What do you notice about the **current** at different points in the circuit?

2. Draw an **energy flow model** to show the forms of energy and the energy transfers in a battery-light bulb circuit. What are the *components* of the system? What *forms of energy* are stored in each component? How is energy *transferred* between components?

3. Explain the difference between the Circuit 1, 2, and 3 in terms of **<u>energy transfers</u>**.

4. The graph below shows the total energy (heat and light) radiated out of the light bulb over time. Use different colors to add a line on the graph for each Circuit 1 and Circuit 3. Note that there are no real numbers here, we are just getting a semi-quantitative comparison.



5. What do you notice about the graph? How does the ***slope*** of the graph relate to the brightness of the bulb?

6. Use the Ammeter in the simulation to measure the current in each of your circuits. Then explore the relationship between current and voltage in each of your three circuits. Record your results on the table below. Then use the electric power equation to calculate the power and energy used of each circuit.

	Voltage	**Current**	**Power**	**Energy used in 2 hours**	
	(Volts)	**(Amps)**	**(Watts)**	**(Watt-hours)**	**(Joules)**
Circuit 1					
Circuit 2					
Circuit 3					

7. How is the amount of energy used related to the **current** in the circuit?

8. How is the amount of energy used related to the **power** in the circuit?

Energy and Power in the Real World

9. A traditional 60 Watt incandescent light bulb uses a tungsten filament that gets hot when electric current runs through it; the hot filament glows and emits radiation. Modern LED lights use semiconductor technology instead of tungsten filaments. An LED that uses 9 Watts of power is just as bright as a traditional 60 Watt bulb.





Your parent is about to buy a 60 Watt incandescent light bulb because they "like the feel of the light." Use your understanding of **energy** and **power** to explain why the 9 Watt LED bulbs are a better choice for your power bill.

10. A Toyota bZ4x electric car has a power rating of 201 horsepower while a Ford F150 Lightning electric pickup truck is rated at 580 horsepower.





Why would a higher horsepower motor be necessary for a large truck compared to a small car? (Hint: Think about the equation for kinetic energy.)